Failure of contiguous-minor closure for Star of David matrices

Principia Math in Progress

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Task 1. Resolve Sun's contiguous-minor conjecture. For $j \geq 1$, Definition 2.12 of [1] sets

$$A^{[j]}_{n,k} = \det[A_{n+i,k+\ell}]_{0 \leq i,\ell < j}.$$

Conjecture 2.15 asserts that $A \in \mathrm{SDR}_\infty$ implies $A^{[j]} \in \mathrm{SDR}_\infty$ for every $j \geq 1$. We refute the assertion already for $j = 2$. We use Sun's source definition: SDR means *Star of David rule*. The added phrase "sign-determined Riordan matrices" in MathDB is inconsistent with that definition. No Riordan-array restriction is imposed in the source conjecture; our result resolves that original assertion, and makes no assertion about a separately restricted class.

For an infinite lower triangular matrix X, with zero entries above the diagonal, put

$$L_{p,r}(n,k;X) = \prod_{i=0}^{r} X_{n+i,k+r-i} \prod_{i=0}^{p-r-1} X_{n+p-i,k+r+i+1},$$

$$R_{p,r}(n,k;X) = \prod_{i=0}^{r} X_{n+p-i,k+p-r+i} \prod_{i=0}^{p-r-1} X_{n+i,k+p-r-i-1}.$$

Definition 1.1 in [1] requires $L_{p,r} = R_{p,r}$ for all $n \geq k \geq 0$, $2 \leq p \leq m-1$, and $0 \leq r \leq p-1$. Membership in SDR_∞ means these identities hold for every $p \geq 2$.

Lemma 1. *Let M be a lower triangular $d \times d$ matrix satisfying all the displayed identities whose row indices lie in $\{0, \ldots, d-1\}$. Then $M \oplus I \in \mathrm{SDR}_\infty$, where I is the infinite identity.*

Proof. Each product uses every row $n, \ldots, n+p$ and every column $k, \ldots, k+p$ exactly once. If $n > k$ and $n + p \geq d$, row $n + p$ belongs to the identity tail. Its only nonzero entry is in column $n + p$, exceeding all the columns in either product. Consequently both products vanish. If $n + p < d$, the assumed finite identities apply.

It remains to handle $n = k$. For $r \geq 1$, the first factor $X_{n,k+r}$ of L is above the diagonal; for $r = 0$, the factor with $i = p - 1$ in its second product is $X_{n+1,k+p} = 0$. In R, if $r \leq p - 2$, its second product contains $X_{n,k+p-r-1} = 0$; if $r = p - 1$, its first product contains $X_{n+1,k+p} = 0$. Thus both sides vanish here too. □

Consider

$$M = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 2 & 1 \end{pmatrix}, \qquad A = M \oplus I.$$

All identities within M hold. To check this without any unbounded computation, cases $n = k$ were proved in the lemma. For $n > k$ and $n + p \leq 4$, the only possibilities are the following:

p	(n,k)	r	L	R
2	$(1,0)$	0	$1 \cdot 0 \cdot 1 = 0$	$1 \cdot 1 \cdot 0 = 0$
2	$(2,0)$	0	$0 \cdot 1 \cdot 1 = 0$	$0 \cdot 1 \cdot 0 = 0$
2	$(2,1)$	0	$1 \cdot 0 \cdot 1 = 0$	$2 \cdot 1 \cdot 0 = 0$
3	$(1,0)$	0	$0 \quad (M_{2,3} = 0)$	$0 \quad (M_{1,2} = 0)$
3	$(1,0)$	1	$0 \quad (M_{2,0} = 0)$	$0 \quad (M_{2,0} = 0)$

For $p = 2, r = 1$, the two sides are the $r = 0$ sides interchanged. For $p = 3, r = 2$, they are the $r = 0$ sides interchanged. There are no further cases with $n > k$. The lemma proves $A \in \mathrm{SDR}_\infty$.

Theorem 1. *For the infinite matrix A above, $A \in \mathrm{SDR}_\infty$ but $A^{[2]} \notin \mathrm{SDR}_3$, and hence $A^{[2]} \notin \mathrm{SDR}_\infty$.*

Proof. Put $C = A^{[2]}$. Its six entries needed for the order-three rule are

$$C_{1,0} = \det \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = 1, \qquad C_{3,1} = \det \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = -1, \qquad C_{2,2} = 1,$$

$$C_{3,2} = \det \begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix} = 2, \qquad C_{1,1} = 1, \qquad C_{2,0} = \det \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = 0.$$

Consequently the identity at $(p, n, k, r) = (2, 1, 0, 0)$ fails:

$$L = C_{1,0}C_{3,1}C_{2,2} = -1 \neq 0 = C_{3,2}C_{1,1}C_{2,0} = R.$$

Every entry used is a minor of the displayed five-by-five block; the extension lemma verifies the stronger infinite-order hypothesis on the input array, completing the counterexample. □

Verification and scope. The supplied script checks all 25 finite identities, the displayed matrix operations, and the exact failed identity. It also checks a larger truncation as a diagnostic. Lemma 1 supplies the proof for the entire infinite matrix; the diagnostic is not used to infer an infinite assertion. Zero entries are allowed in Definition 1.1 and in the selected conjecture. The source explicitly imposes a nonzero-entry hypothesis in other theorems, including Theorem 2.10, but not here. Remark 2.16 in the source gives an example with input only in order three. That example does not meet the infinite-order hypothesis here. Our input does. The same input also refutes the different inverse-closure conjecture, treated as MathDB 317353; the present determinant calculation resolves the separately stated minor operation.

Task 2. Minimize the minor counterexample. Find the smallest leading block supporting a failure of contiguous two-by-two-minor closure under the infinite-order hypothesis.

Task 3. Find sufficient support conditions. Characterize zero patterns for which every contiguous-minor transform preserves the Star of David identities.

Task 4. Study iterated minor transforms. Determine how failures propagate under repeated contiguous-minor transforms, starting from sparse unit lower triangular arrays.

Task 5. Separate closure properties. Construct arrays that distinguish inverse closure from contiguous-minor closure, and determine any implications between these two properties.

References

[1] Y. Sun, *The Star of David rule*, Linear Algebra Appl. **429** (2008), 1954–1961, doi:10.1016/j.laa.2008.05.027; arXiv:0805.1277v1. Source references in this note use the actual seven-page arXiv PDF.